ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ASSIGNMENT – 3

Apply Dimensionality Reduction Using PCA

Dataset: Scikit-learn Diabetes Dataset

1. Aim

To apply Principal Component Analysis (PCA) for reducing the dimensionality of the Diabetes dataset while retaining as much of the original information as possible.

2. Objectives

- To understand dimensionality reduction and the purpose of PCA.
- To standardize the input features before PCA.
- To transform the original 10-dimensional feature space into principal components.
- To analyze explained and cumulative variance.
- To visualize the reduced data using the first two principal components.

3. Dataset Description

The Scikit-learn Diabetes dataset contains 442 observations and 10 independent features: age, sex, bmi, bp, s1, s2, s3, s4, s5 and s6. For regression experiments, the continuous target is used directly. For classification experiments, the continuous target was converted into two classes using the median value of 140.5: Lower Progression and Higher Progression.

Property	Value
Samples	442
Original features	10
Target	Continuous target
Technique	PCA

4. Theory

PCA is a linear dimensionality-reduction technique that transforms a set of possibly correlated variables into a smaller set of uncorrelated variables called principal components. It is widely used for visualization, noise reduction, feature compression and simplifying high-dimensional datasets.

The principal components are ordered according to the amount of variance they explain. The first principal component captures the maximum possible variance subject to being orthogonal to later components. The second captures the largest remaining variance, and this process continues for subsequent components.

Before applying PCA, feature scaling is important when variables are measured on different scales. Standardization gives each feature approximately zero mean and unit variance, preventing variables with larger numerical magnitudes from dominating the principal components.

PCA can be understood geometrically as finding a new coordinate system for the data. The new axes are directions in feature space that capture decreasing amounts of variation. Because the components are orthogonal, they do not contain the same linear direction of variation.

The explained variance ratio of a principal component indicates the proportion of the total variance represented by that component. The cumulative explained variance is obtained by adding the explained variance ratios of the components selected so far.

A practical advantage of PCA is that many original variables can sometimes be represented by substantially fewer components. However, PCA components are combinations of the original variables, so they can be less directly interpretable than the original features.

In this experiment, PCA is applied to all ten standardized independent variables. The target is not included in the transformation because the objective is dimensionality reduction of the feature space rather than supervised prediction.

The first two components can be plotted to obtain a two-dimensional representation of the original data. This visualization is useful for understanding broad patterns and possible structure, although reducing to two dimensions may discard some information.

Mathematically, PCA can be obtained from the covariance or correlation structure of the standardized features. Eigenvectors determine the directions of the principal components, while the corresponding eigenvalues indicate the amount of variance associated with those directions.

The transformed data is obtained by projecting each standardized observation onto the selected principal-component directions. If X is the standardized feature matrix and W contains the component directions, the transformed representation can be written conceptually as $Z = XW$.

The choice of the number of components is a trade-off between compression and information retention. A common approach is to choose enough components to retain a desired percentage of variance, such as 90%, 95% or 99%.

The PCA loadings indicate how strongly each original feature contributes to a component. Large positive or negative loadings indicate that the corresponding original variable has a strong influence on that component. The sign of a component is mathematically arbitrary, so the relative pattern of loadings is more important than the sign alone.

In the present experiment, PC1 explains 40.24% of the variance and PC2 explains 14.92%. Together, the first two components retain 55.17% of the total variance. Seven components are required to retain at least 90% variance, while eight are required to retain at least 95%.

5. Methodology

- Load the Diabetes dataset.
- Separate the ten independent features from the target.
- Standardize the feature matrix.
- Fit PCA using all ten components.
- Calculate explained variance and cumulative variance.
- Determine the number of components needed for 90% and 95% variance.
- Transform the data into two principal components for visualization.
- Inspect the PCA loading values.

6. Outputs

6.1 Explained Variance Results

Principal Component	Explained Variance	Explained Variance (%)	Cumulative Variance	Cumulative Variance (%)
1.0	0.4024	40.2421	0.4024	40.2421
2.0	0.1492	14.9232	0.5517	55.1653
3.0	0.1206	12.0597	0.6722	67.2250
4.0	0.0955	9.5548	0.7678	76.7797
5.0	0.0662	6.6218	0.8340	83.4015
6.0	0.0603	6.0272	0.8943	89.4287
7.0	0.0537	5.3657	0.9479	94.7944
8.0	0.0434	4.3368	0.9913	99.1312
9.0	0.0078	0.7832	0.9991	99.9144
10.0	0.0009	0.0856	1.0000	100.0000

6.2 Explained Variance Graph

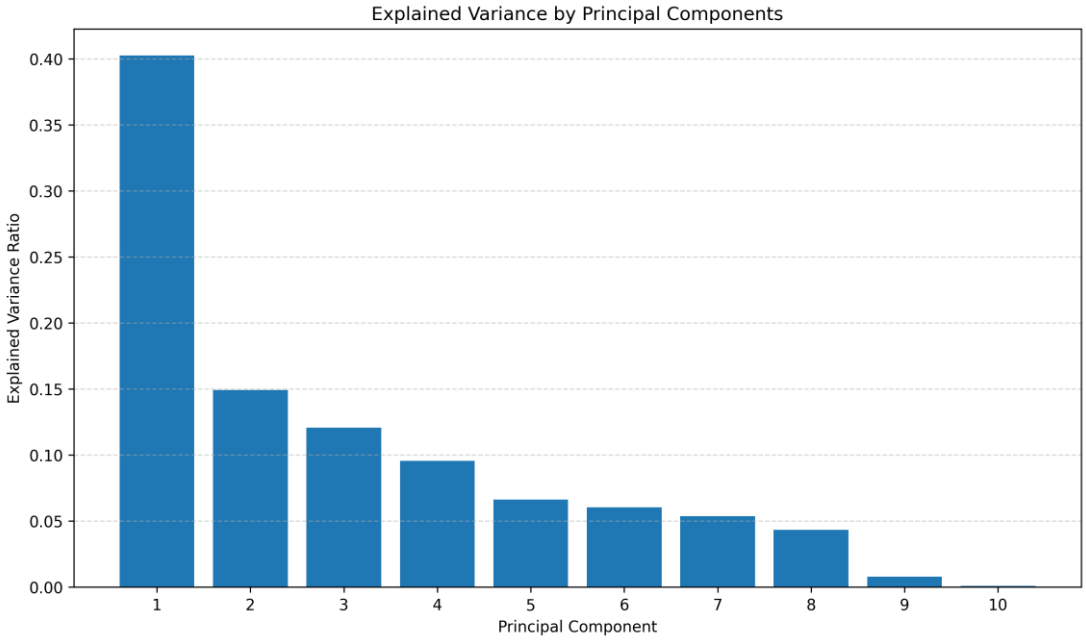


Figure 1: Explained variance of principal components

6.3 Cumulative Variance Graph

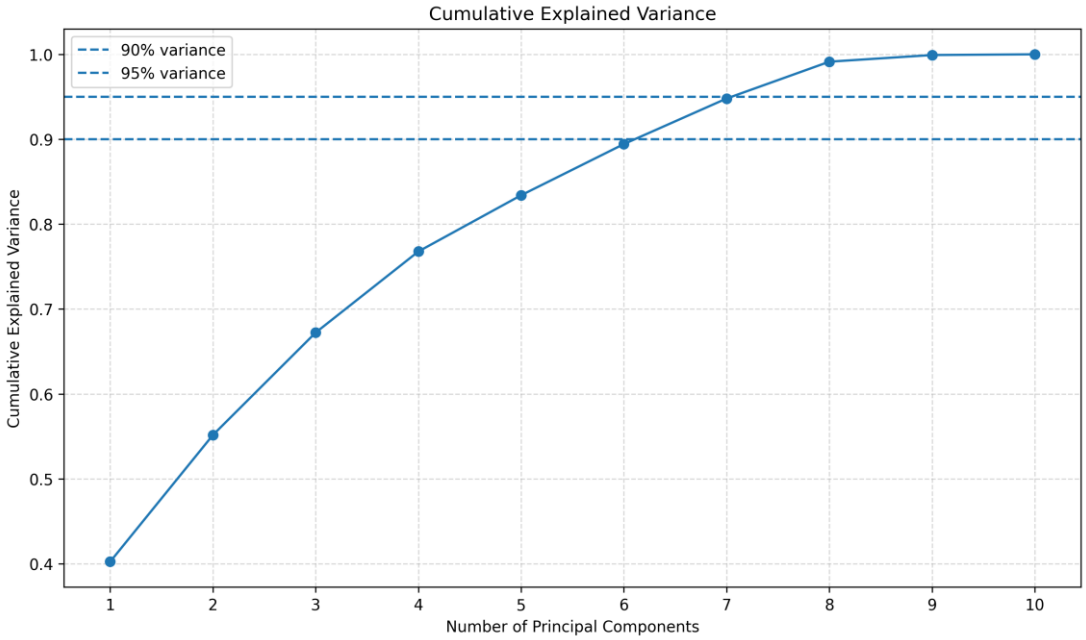


Figure 2: Cumulative explained variance

6.4 PCA 2D Visualization

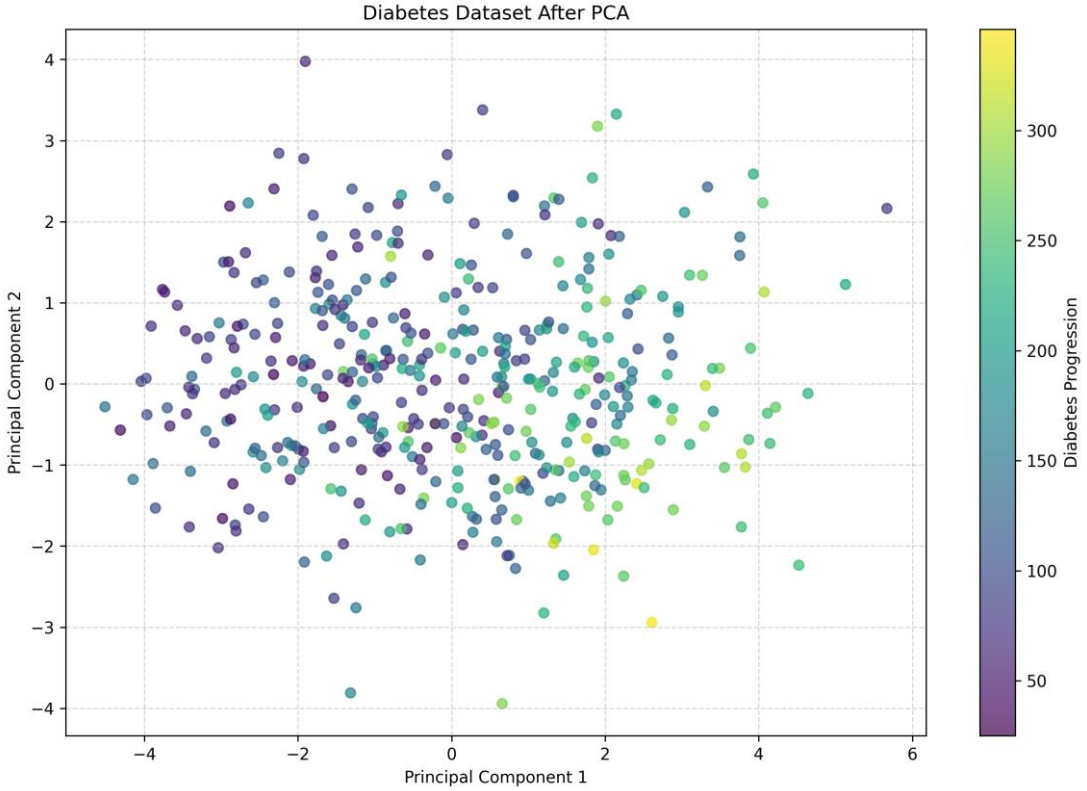


Figure 3: Two-dimensional PCA projection

6.5 PCA Loadings Heatmap

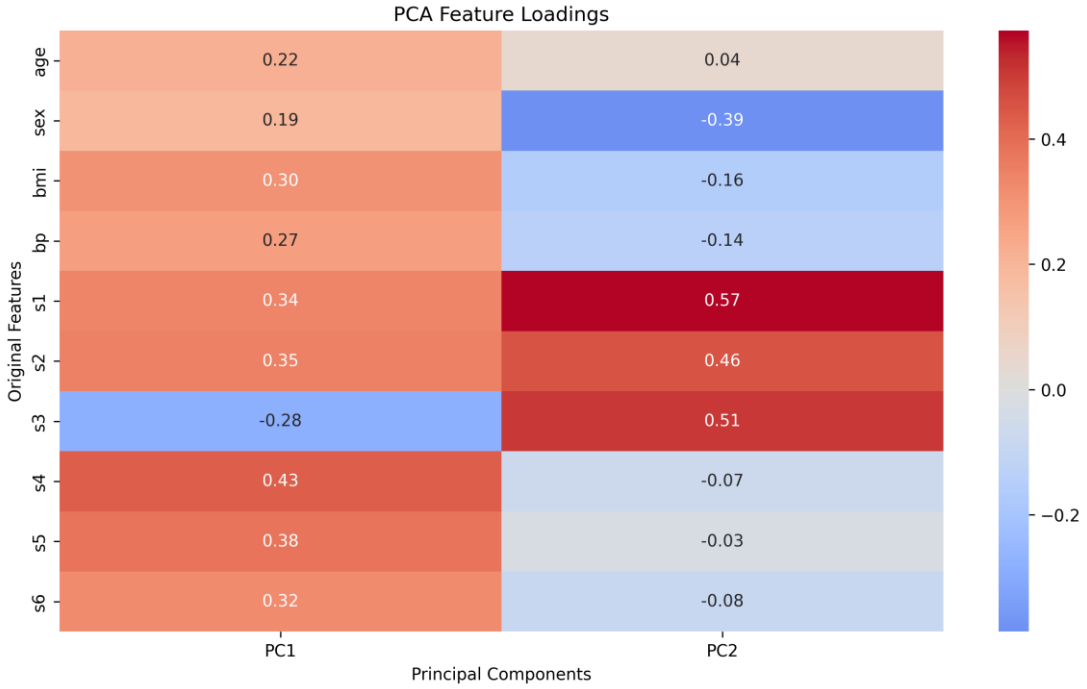


Figure 4: PCA loading heatmap

7. Result

PCA was successfully applied to the 10-dimensional Diabetes feature space. PC1 explains 40.24% and PC2 explains 14.92% of the variance, giving 55.17% cumulative variance for the first two components. Seven components retain at least 90% variance and eight retain at least 95%.

8. Observations

- PC1 captures the largest proportion of variation.
- The amount of explained variance generally decreases for later components.
- The first two components provide a compact visualization but retain only 55.17% of total variance.
- Seven components are sufficient for at least 90% variance retention.
- The loading matrix shows how original features contribute to PC1 and PC2.

9. Conclusion

The experiment demonstrates that PCA can reduce the dimensionality of the Diabetes dataset while retaining a large proportion of its variance. The explained-variance analysis provides a quantitative basis for selecting the number of components, while the two-dimensional projection makes the high-dimensional dataset easier to visualize.